33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F21-SR-001
415V CIL4 SWITCHROOM

F21-TR-007
TRANSFORMER ROOM

F21-TR-007
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF

SWR EARTH BAR
Tx EARTH BAR
NOTE 3
4000/ 1A
2.5VA
5P20

1600A
4 POLE
F21-GE-007
415V
1000kVA

F21-SW-007
CIL4 415V MOTOR CONTROL CENTRE No.2

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
5VA
Cl 1
4000/ 5A
30VA
Cl0.5
1600/ 5A
5VA
Cl 1
1600/ 1A
2.5VA
5P20

TEST
PM
AFD
ACB
00F-08
01F-08
4000A
1600A
20A
63A
2A
32A
2A
63A
240VAC SPRING
CHARGE SUPPLY
ATS
24VDC CONTROL POWER SUPPLY
NOTE 7
NOTE 4
NOTE 6
125A
160A
SD

CONTINUED ON SHT 2
2000-F21-SLD-EL-10252

EMERGENCY 415V 3NE~ 50Hz 4000A 65kA for 1sec

06F-05
05F-05
03F-05
05R-05
05R-05
INC
06R-11
06R-18
INC
02F-04
02F-11

630A
250A
90A
EOL
xxxA
xxxA
xx / xxA
xxVA
xPxx

E-SHUT
F21SR001-LCS
NOTE 12

RTD
M
3~
250kW
F21-AG-415
CIL4
TANK 5 AGITATOR
2000-F21-SCM-EL-10259
F21-AG-416
CIL4
TANK 6 AGITATOR
2000-F21-SCM-EL-10260
F21-AG-417
CIL4
TANK 7 AGITATOR
2000-F21-SCM-EL-10261
F21-AG-418
CIL4
TANK 8 AGITATOR
2000-F21-SCM-EL-10262
F21-AG-419
CIL4
TANK 9 AGITATOR

75kW
F21-SC-415A
CIL4 TANK 5
INTERTANK SCREEN 1
2000-F21-SCM-EL-10270
F21-SC-415B
CIL4 TANK 5
INTERTANK SCREEN 2
2000-F21-SCM-EL-10271
F21-SC-416A
CIL4 TANK 6
INTERTANK SCREEN 1
2000-F21-SCM-EL-10272
F21-SC-416B
CIL4 TANK 6
INTERTANK SCREEN 2
2000-F21-SCM-EL-10273

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Date 19/11/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER TO THAT EFFECT. CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F21-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 12)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES	REV	DATE	DESCRIPTION
2000-F21-SCM-EL-10254	F21-SW-007 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM			
2000-F21-SCM-EL-10253	F21-SW-007 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM			
2000-F21-SCM-EL-10251	F21-SW-007 415V MCC INCOMER SCHEMATIC DIAGRAM			
2000-F21-DRG-EL-10101	F21-SR-001 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT			
2000-F21-BLK-IC-10001	F21-SR-001 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM			
2000-F21-SLD-EL-10252	F21-SW-007 415V MCC SINGLE LINE DIAGRAM SHEET 2			
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5	1	20OCT25	FAT REDLINES INCORPORATED
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX	0	28JAN25	APPROVED FOR CONSTRUCTION

[redacted] | DRAW | CHECK | DESIGN | TECH APP | PROJ APP

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: FLOTATION TAILINGS CIL4
F21-SW-007 415V MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F21-SLD-EL-10251	1

O:\CAD PROJECTS\23S NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F21 - FLOTATION TAILINGS CIL 4\2000-F21-SLD-EL-10251DWG 06NOV25 11:07AM HOTSEAT04